

Reinforcement Learning for Full-Body Humanoid Walking and Soccer Kicking in Isaac Lab

Shanshan Zhu, Aaditya Voruganti, Prithvi Teekaa Murali
CS 498 Robotics Team Project
University of Illinois Urbana-Champaign

Abstract—This project investigates reinforcement learning (RL) for humanoid locomotion and soccer-style kicking using the Booster K1 humanoid robot in Isaac Lab. Our goal was to train a stable full-body walking policy and extend it toward dynamic ball interaction and kicking behaviors. We implemented a custom RL environment using Isaac Lab and RSL-RL Proximal Policy Optimization (PPO) training, incorporating full-body joint control, reward shaping, motion stabilization constraints, and ball interaction objectives.

Our final system achieved stable forward locomotion with improved balance and directional control in simulation. We additionally explored kicking-oriented reward design and ball interaction training, but stable and repeatable soccer kicking remained challenging due to training instability, sparse interaction rewards, and limited time for policy refinement. This report summarizes our environment design, implementation pipeline, experimental results, debugging process, and lessons learned from humanoid RL.

Index Terms—humanoid robotics, reinforcement learning, Isaac Lab, locomotion, PPO, soccer robotics

I. INTRODUCTION AND MOTIVATION

Humanoid locomotion remains one of the most difficult problems in robotics because the robot must maintain balance while coordinating many joints under complex contact dynamics. The problem becomes even harder when the robot interacts with external objects such as a soccer ball. A successful soccer behavior requires stable walking, accurate timing, coordinated leg motion, and recovery after contact. Small control errors can easily destabilize the entire body.

This project explored reinforcement learning (RL) for full-body humanoid walking and soccer-style kicking using the Booster K1 humanoid robot in Isaac Lab. Our initial goal was to train a policy that could both walk stably and generate controlled kicking behaviors in simulation. We focused on simulation because RL for humanoid systems requires large amounts of interaction data, and Isaac Lab provides massively parallel GPU-based simulation environments that make this training process feasible.

At the beginning of the project, we expected that kicking could be developed by extending an already stable walking policy with additional ball interaction rewards. Early experiments suggested that this approach might work because the humanoid could already maintain balance during forward locomotion. Yet the later training process showed that locomotion and kicking compete with each other in important ways. Policies that produced stable walking often avoided aggressive contact with the ball, while policies that attempted stronger

kicking motions frequently lost balance or collapsed during training. This observation became one of the central findings of the project.

Our work therefore shifted toward understanding how reward design, motion stability, and contact dynamics interact during humanoid RL. The final system achieved stable full-body walking and demonstrated several emerging kicking-like behaviors, including directed leg swings and intermittent ball contact. However, robust and repeatable soccer kicking was not consistently achieved within the project timeline.

II. RELATED WORK AND BACKGROUND

Recent progress in humanoid robotics has demonstrated that RL can produce highly dynamic locomotion behaviors when combined with large-scale graphics processing unit (GPU) simulation. Physics simulation platforms such as MuJoCo [1], the DeepMind Control Suite [2], Isaac Gym [3], and Isaac Lab [4] have significantly accelerated research in robot learning and continuous control. In particular, Isaac Gym and Isaac Lab support massively parallel GPU-based simulation, enabling thousands of humanoid environments to be executed simultaneously during RL training. Large-scale GPU simulation has substantially improved locomotion training efficiency. Rudin et al. [5] demonstrated that massively parallel RL can train locomotion policies within minutes, while Hwangbo et al. [6] showed that agile and dynamic motor skills can emerge through sim-to-real locomotion training pipelines.

Many modern locomotion systems rely on Proximal Policy Optimization (PPO) [7] because it provides relatively stable optimization for high-dimensional continuous control problems. PPO has been widely adopted in humanoid locomotion, quadruped control, and dexterous manipulation tasks due to its stability and implementation simplicity. Alternative RL algorithms such as Soft Actor-Critic (SAC) [8] have also demonstrated strong performance in continuous-control settings, although PPO remains particularly common in large-scale locomotion training pipelines.

Our project was additionally influenced by open-source humanoid robotics systems from Booster Robotics [9] and RoboCup humanoid soccer environments [10]. These systems highlight the importance of stable whole-body control during dynamic interaction tasks such as walking, turning, and ball contact.

Compared with standard locomotion benchmarks, humanoid soccer introduces additional difficulties because successful ball

interactions are sparse during early training. The robot must coordinate aggressive leg swings while maintaining upright balance and recovering after contact. Prior work has also explored dynamic humanoid soccer behaviors and coordinated contact interaction in competitive robotic settings [11]. During our experiments, we repeatedly observed that policies optimized primarily for stability often converged toward conservative walking behaviors instead of dynamic kicking motions. This tradeoff is related to broader challenges in adaptive locomotion and robust motor control discussed in prior work such as RMA [12].

Previous work has also shown that motion priors and imitation learning can improve stability in complex humanoid tasks. DeepMimic [13] demonstrated that motion imitation can help RL produce coordinated full-body behaviors from reference motion trajectories. These approaches often improve motion naturalness and stability compared with pure reward-shaping methods. Unlike imitation-based systems, our project explored whether soccer-oriented interaction behaviors could emerge primarily from RL and reward shaping alone.

III. METHOD AND APPROACH

The overall system consisted of a custom Isaac Lab environment, a PPO-based RL policy, and a reward structure designed to encourage stable locomotion and ball interaction. The Booster K1 humanoid robot served as the primary simulated platform.

The observation space included robot proprioceptive information and environment state variables. Observations contained base linear velocity, base angular velocity, projected gravity, joint positions, joint velocities, previous actions, and commanded walking velocity. During kicking experiments, ball-relative observations were also introduced so that the policy could estimate the position of the ball relative to the humanoid body.

The action space controlled full-body target joint offsets. Both lower-body and upper-body joints were included because arm motion contributed to stabilization during locomotion and kicking attempts. Actions were applied directly through Isaac Sim physics simulation at every environment step.

The RL pipeline used PPO [7] with massively parallel simulation environments. Thousands of simulation instances were executed simultaneously on the GPU to accelerate policy learning.

The PPO policy objective is defined as:

$$L^{\text{PPO}}(\theta) = \mathbb{E}_t \left[\min(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t) \right] \quad (1)$$

where $r_t(\theta)$ represents the ratio between the updated and previous policy probabilities, $\hat{A}_t$ is the estimated advantage function, and ϵ is the PPO clipping parameter.

Reward design became one of the most important parts of the project. The locomotion rewards encouraged forward velocity tracking, heading stability, upright posture, smooth actions, and low lateral drift. Penalty terms discouraged unstable yaw rotation, excessive torque usage, and abrupt motion changes.

The overall reward function combined locomotion and interaction objectives:

$$R_t = w_v r_{\text{velocity}} + w_o r_{\text{orientation}} + w_s r_{\text{smoothness}} + w_b r_{\text{ball}} - w_p r_{\text{penalty}} \quad (2)$$

where the reward terms encourage stable locomotion, upright posture, smooth motion, and soccer-oriented ball interaction while penalizing unstable motion and excessive torque usage.

Once stable walking emerged, additional rewards were introduced for ball interaction. These included ball contact rewards, forward ball velocity rewards, and ball displacement rewards.

At first, we expected that stronger ball rewards would naturally encourage kicking. The training results showed a more complicated pattern. Increasing ball interaction rewards sometimes produced larger leg swings and more aggressive motion toward the ball. Yet these same reward changes also destabilized locomotion. Policies often regressed toward shorter episode lengths or unstable movement patterns after reward adjustments.

Figure 1 illustrates the overall RL pipeline used for humanoid locomotion and soccer-oriented interaction training in Isaac Lab.

IV. IMPLEMENTATION DETAILS

A. Software Stack

The project was implemented using Isaac Lab [4], Isaac Sim, the RSL-RL training framework [14], and PyTorch. Training was performed on NVIDIA RTX 4090 GPUs using massively parallel simulation environments.

The RL pipeline relied on GPU-accelerated PPO optimization with thousands of parallel humanoid environments executed simultaneously during training. This large-scale simulation setup significantly improved data collection efficiency and reduced policy training time.

B. Custom Isaac Lab Environment

We created a custom full-body locomotion environment for the Booster K1 humanoid robot. The environment included lower-body locomotion joints, upper-body arm joints, contact sensors, reward functions, and termination conditions. The policy controlled approximately twenty joints simultaneously, which made the control problem substantially more difficult than standard leg-only locomotion benchmarks.

Several environment modifications were introduced during development and debugging. Joint initialization offsets were adjusted to improve early-stage balance stability. Reward scales were repeatedly tuned to reduce excessive lateral drift and yaw rotation. Contact thresholds for ball interaction were also modified because even small threshold changes sometimes altered the learned behavior significantly.

The environment additionally incorporated soccer-oriented interaction objectives, including ball contact rewards, forward ball velocity rewards, and ball displacement rewards. These reward terms were combined with locomotion stabilization

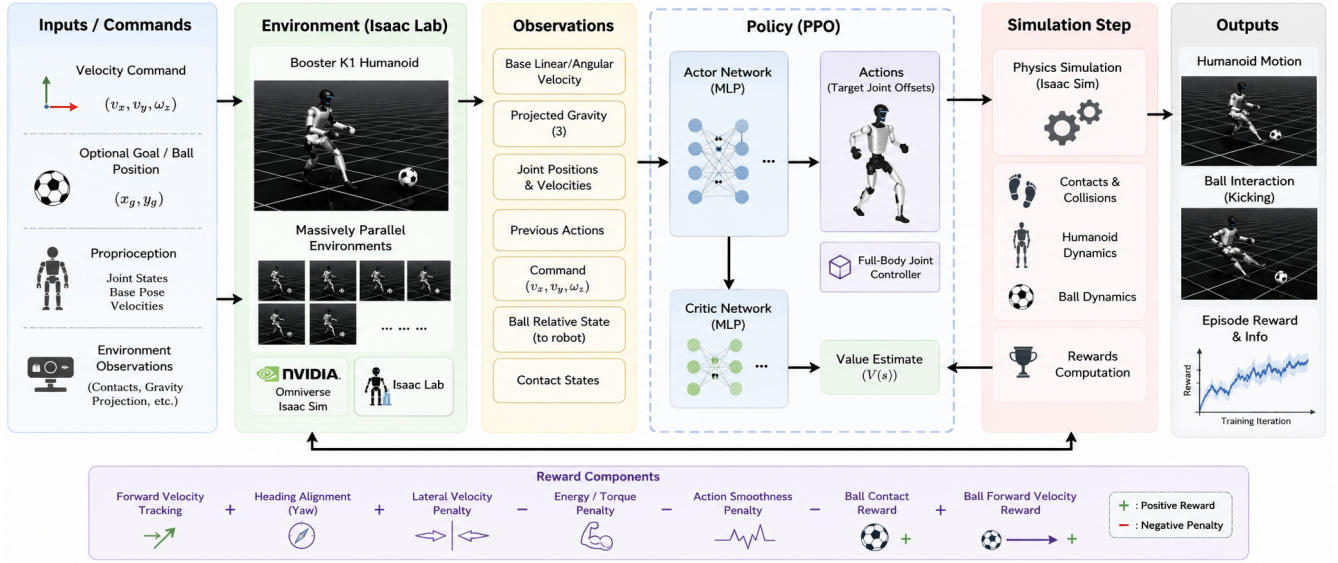


Fig. 1. Overview of the humanoid RL pipeline used in Isaac Lab. The system combines massively parallel simulation, PPO-based policy learning, full-body humanoid control, and reward shaping for locomotion and soccer-oriented interaction behaviors.

rewards to encourage coordinated full-body motion while maintaining upright balance.

C. Training and Debugging Challenges

The training process revealed several engineering difficulties that were not obvious at the beginning of the project. PPO batch statistics often differed from deterministic rollout behavior. In some cases, deterministic evaluations produced visible kicking motions while average PPO training rewards suggested that ball interaction had disappeared.

This mismatch complicated debugging because the same policy could appear successful during isolated rollouts while still converging toward conservative locomotion behaviors during large-scale optimization. Reward interactions also proved highly sensitive. Small modifications to ball interaction rewards sometimes destabilized previously learned locomotion behaviors and caused shorter episode lengths or unstable gait patterns.

Another major challenge involved sparse rewards during kicking experiments. Successful ball contact events occurred infrequently during early training, which limited the amount of useful feedback available to the policy. As a result, locomotion and interaction objectives frequently competed with each other during optimization.

V. EXPERIMENTS AND RESULTS

The experiments focused on two major goals: stable full-body locomotion and soccer-oriented ball interaction. Early training runs concentrated entirely on walking stability before kicking-oriented rewards were introduced.

Figure 2 summarizes representative PPO training dynamics observed during locomotion and soccer-oriented interaction learning.

TABLE I
SUMMARY OF EXPERIMENTAL OUTCOMES

Evaluation Component	Result
Stable forward locomotion	Achieved
Long-horizon balance maintenance	Achieved
Full-body coordination	Achieved
Consistent ball interaction	Partial success
Repeatable soccer kicking	Not achieved
Training convergence stability	Moderate
Primary failure mode	Walking-only local optimum

The training curves revealed clear improvements in locomotion stability during PPO optimization. Mean episode reward increased steadily during training, while episode length also improved substantially, indicating that the humanoid learned more stable full-body walking behaviors over time.

Forward velocity tracking rewards became increasingly consistent as the policy learned to maintain heading stability and reduce lateral drift. In contrast, ball interaction rewards remained significantly more unstable throughout training. Although some policies occasionally generated forward ball motion and kicking-like leg swings, the associated reward terms fluctuated heavily because successful contacts remained sparse and difficult to stabilize.

These results suggest that locomotion learning and soccer interaction objectives compete with each other during optimization. Policies optimized primarily for balance often converged toward conservative walking behaviors, while stronger interaction rewards frequently destabilized previously learned locomotion policies.

The final locomotion policy achieved stable forward walking in simulation. The humanoid maintained upright posture for extended periods and demonstrated smoother full-body coordination compared with earlier training runs. Arm motion also

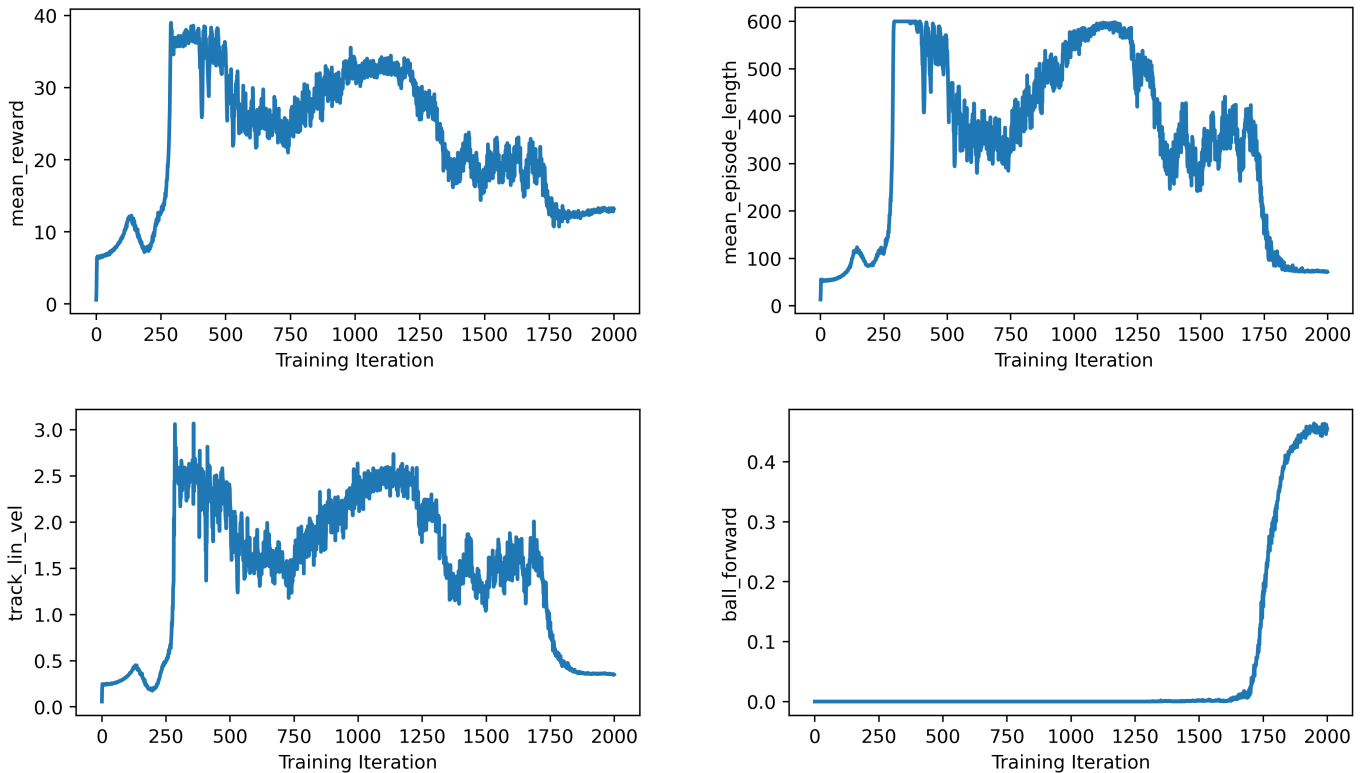


Fig. 2. Representative PPO training dynamics. Top-left: mean PPO reward during training. Top-right: mean episode length indicating locomotion stability. Bottom-left: forward velocity tracking reward. Bottom-right: forward ball interaction reward during soccer-oriented training.

became more natural as the policy learned to stabilize the torso during locomotion.

The kicking experiments produced mixed results. Several policies demonstrated clear kicking-like motions, including directed leg swings toward the ball and intermittent forward ball motion. In some rollouts, the robot successfully contacted the ball and generated visible displacement.

At the same time, the kicking behavior remained inconsistent. Some episodes still resembled walking combined with accidental ball contact rather than fully controlled kicking. Stronger kicking attempts frequently reduced balance stability, and the robot sometimes reverted to conservative walking-only behaviors during longer PPO training runs.

These observations changed our understanding of the problem. At the beginning of the project, we assumed that kicking could be added as a relatively small extension to locomotion training. The experiments instead suggested that stable kicking may require curriculum learning, staged training procedures, stronger motion priors, or imitation-based initialization strategies.

VI. DISCUSSION AND FUTURE WORK

The experiments demonstrated that RL can successfully produce stable full-body humanoid locomotion in Isaac Lab using massively parallel simulation and PPO optimization. The learned policies achieved stable forward walking, improved

heading control, and smoother full-body coordination compared with early-stage training runs. These results suggest that large-scale GPU simulation can effectively support complex humanoid locomotion learning even under high-dimensional control settings.

The main challenge came from the interaction between locomotion stability and ball interaction rewards. Policies that preserved balance tended to avoid aggressive kicking behavior, while policies that attempted stronger kicks often destabilized quickly. This issue became especially visible during longer training runs where the policy gradually converged toward safer locomotion strategies.

Another challenge involved sparse rewards. Ball contact events occurred infrequently during early training, so the policy often received limited feedback about successful interaction. Small reward modifications sometimes produced unexpectedly large behavioral changes, which made training difficult to debug.

The experiments also revealed differences between deterministic rollout behavior and PPO batch statistics. In some cases, deterministic rollouts showed visible kicking-like behavior while average PPO metrics suggested that ball interaction performance had decreased. This mismatch complicated policy evaluation and debugging.

If more time were available, several improvements would likely help. Curriculum learning could gradually introduce kicking behavior after stable locomotion is achieved. Motion

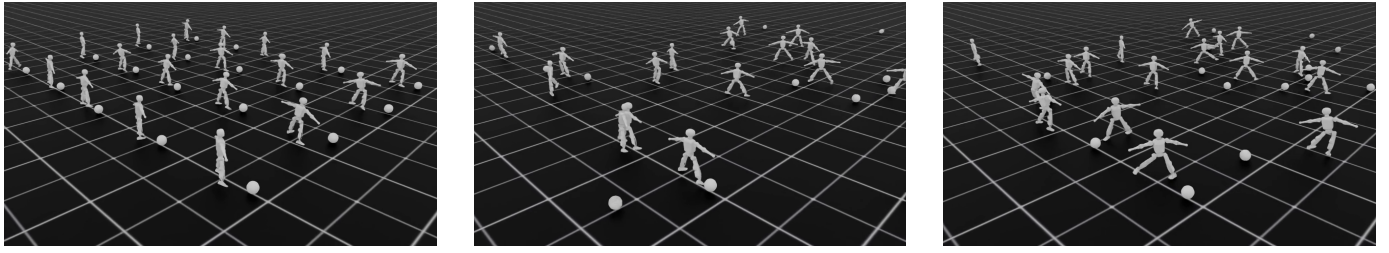


Fig. 3. Qualitative humanoid soccer behaviors learned in Isaac Lab. (a) Stable full-body locomotion across massively parallel simulation environments. (b) Ball approach and interaction behaviors emerging during reinforcement learning training. (c) Kicking-oriented motions and dynamic leg swing behaviors observed during policy rollout.

imitation [13] could provide stronger priors for coordinated leg swings. Additional contact-aware rewards, structured ball interaction objectives, and domain randomization would also likely improve stability and robustness before sim-to-real deployment.

VII. TEAM MEMBER CONTRIBUTIONS

- **Shanshan Zhu:** Designed RL environments, implemented PPO training pipelines, performed reward engineering, conducted training and debugging experiments for both walking and kicking policies, generated evaluation figures, and wrote the report.
- **Aaditya Voruganti:** Contributed to humanoid walking policy development, simulator setup, locomotion evaluation, and experiment analysis.
- **Prithvi Teekaa Murali:** Contributed to kicking-policy development, interaction testing, presentation preparation, and project documentation.

VIII. CONCLUSION

In this project, we developed a RL pipeline for full-body humanoid locomotion and soccer-oriented interaction using the Booster K1 humanoid robot in Isaac Lab.

Our system successfully achieved stable forward locomotion with coordinated full-body movement and improved balance stability. We additionally explored soccer-oriented interaction rewards and observed emerging kicking-like behaviors during training. However, robust and repeatable soccer kicking was not consistently achieved due to training instability, sparse rewards, and the difficulty of balancing locomotion and interaction objectives simultaneously.

Overall, this project provided valuable experience in humanoid RL, reward design, large-scale simulation training, and debugging complex contact-rich robotic behaviors.

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